# SQL Server Assessment Results

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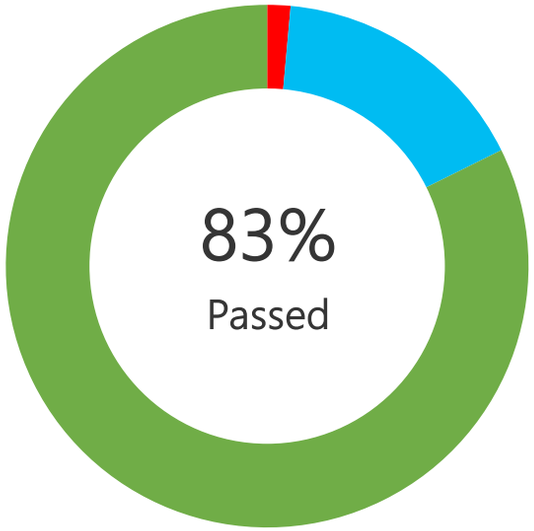
Microsoft

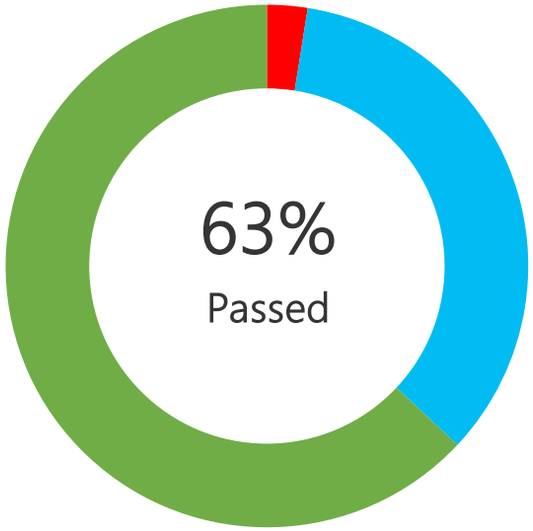
# Executive Summary

1. What went well:
    - Operations and Monitoring
2. What went poorly:
    - Security and Compliance
3. How to improve:
    - Review the build number of the SQL Server instance.




- 12 High Priority
- 136 Low Priority
- 689 Passed Checks

# Security and Compliance

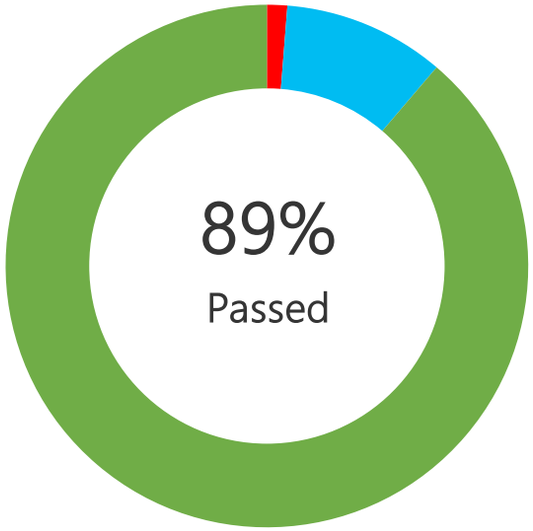



- 5 High Priority
- 69 Low Priority
- 126 Passed Checks

## Highest Priority Recommendations

- Disable SMB1 protocol to protect your network from security risks such as man-in-the-middle-attack
- Configure and Enforce the Setting "Windows Firewall: Domain: Firewall state" via GPO
- Review remote connections using NTLM Authentication
- Consider disabling the xp_cmdshell extended stored procedure.
- Configure the Setting "Network security: LAN Manager authentication level" and Enforce via GPO
- Enable and Enforce "Microsoft network server: Digitally sign communications (if client agrees)" via GPO
- Enforce Security Setting "Accounts: Limit local account use of blank passwords to console logon only" with GPO
- Define and Enforce Setting "Back up files and directories" via GPO for Accounts Which Need This Level of Access
- Configure Authorized User List for Setting "Restore files and directories" and Enforce via GPO
- Avoid using and account with local administrator privileges to run the SQL Server service.

# Operations and Monitoring




1 High Priority

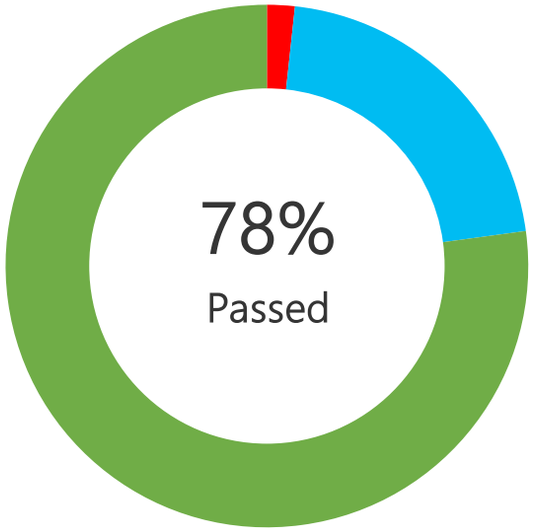
8 Low Priority

71 Passed Checks

## Highest Priority Recommendations

- Create a schedule to run DBCC CHECKDB regularly.
- Review the enabled SQL Server trace flags to see if they are still valid.
- Investigate the Application Event Log: Perflib:2003 – The configuration information of the performance library does not match the trusted performance library information.
- Configure Auditing for Account Management: Security Group Management
- Configure Auditing for Account Management: User Account Management
- Configure Auditing for Object Access: Audit File Share
- Configure Auditing for Object Access: Audit Other Object Access Events
- Configure Auditing for Object Access: Removable Storage
- Review the user objects in the master database.

# Performance and Scalability




3 High Priority

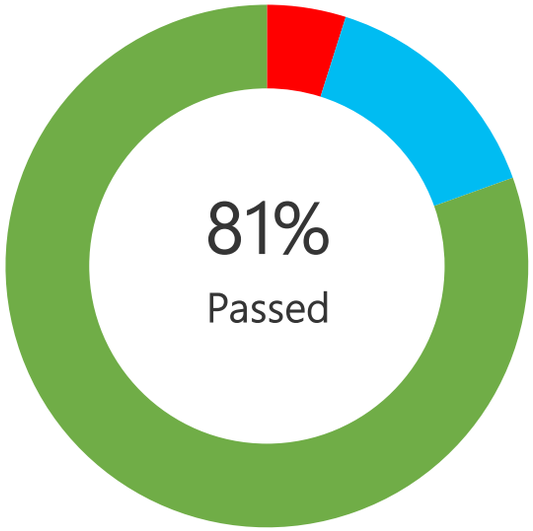
37 Low Priority

135 Passed Checks

## Highest Priority Recommendations

- Investigate slow disk read times.
- Investigate slow disk write times.
- Review and update statistics that have had significant changes since last update
- Consider to change or remove Indexes with an index key larger than the recommended size.
- Review the file sizes for the tempdb data files.
- Defragment indexes to improve performance.
- Increase columnstore compression gains.
- Consider deleting duplicate indexes on tables and/or indexed views in databases.
- Consider creating supporting indexes for tables with foreign keys.
- Consider creating clustered indexes for database tables that do not have them.

# Upgrade, Migration and Deployment




2 High Priority

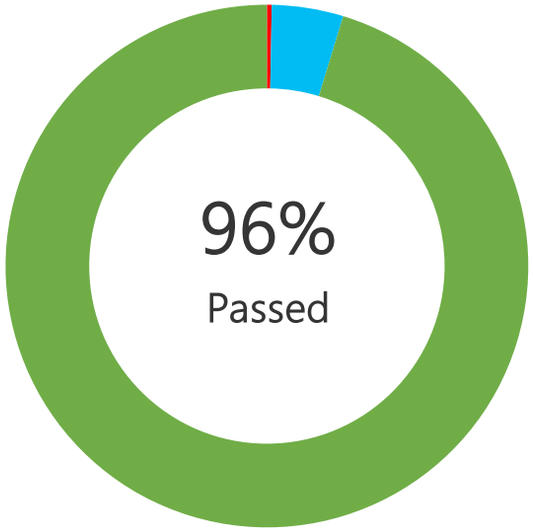
6 Low Priority

33 Passed Checks

## Highest Priority Recommendations

- Review the build number of the SQL Server instance.
- Avoid using deprecated database SET options
- Install the latest service pack or hotfix for the SQL Server.
- Avoid using timestamp columns with null values as it is deprecated
- Verify the compatibility of third party modules loaded into SQL Server address space in case of upgrade
- Reduce the number of rows in backup or restore history tables as they might make upgrade appear to hang
- Review the third-party OS Virtualization that SQL Server is running on.
- Consider enable server network protocol Named Pipes on side by side migration

# Availability and Business Continuity




1 High Priority

15 Low Priority

323 Passed Checks

## Highest Priority Recommendations

- Change the recovery model to FULL or BULK_LOGGED.
- Investigate the SQL Error Log 3041: BACKUP failed to complete the command.
- Review the SQL Server Instance configuration setting for recovery interval and change it to the default value.
- Set Sysadmins to use master as the default database
- Investigate SQL Error Log message 17806 – SSPI handshake failed while establishing a connection with integrated security.
- Increase free space on system drives
- Some jobs or job steps are in failed status
- Investigate SQL Error Log message 18056 – The client was unable to reuse a session.
- Review the non-default Scan for Startup Procs setting on Microsoft SQL Server.
- Review the SQL Server instance configuration setting for remote admin connections and configure the value to match your company policy.